

Collaboration within architecture and structure based on BIM platforms

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Abstract — Building Information Modelling (BIM) is a methodology focused on the centralization and sharing of the project information among all professionals involved, supported on the generation and manipulation of a three-dimensional (3D) digital BIM model. This methodology allows adequate coordination and team collaboration, avoiding conflicts between specialties and data omissions. In the present study, the benefits and limitations identified within the collaboration and integration between structural and architectural professionals, using BIM platforms were analyzed. A structural design case, elaborated within a BIM environment was developed, using the Revit modelling and structural calculation software Robot. The main remarks concerns the fundamental aspects of information sharing, interoperability, and reinforcement detailing. Despite the limitations found, essentially related to the software interoperability, the BIM approach presents significant advantages in structural design.

Keywords - BIM; structural design; interoperability; integration; collaboration.

I. INTRODUCTION

Currently, building design present volume, complexity, and multidisciplinary integration, which by the traditional method based on drawing, becomes difficult to manage in time and control of costs. The fundamental concept of BIM methodology, referring to a hierarchical system of building components in the composition of a digital model, was first mentioned by Eastman [1], in 1975, through the designation Building Description System (BDS). Later, as referred by Succar [2], in 2009, Building Information Modelling (BIM) was adopted, as the most common designation for this new way of approaching design representation. The elaboration of a building project, involving all disciplines and the required associated task, using the available BIM software, is provided in the form of a digital parametric model, containing the geometric details necessary for its realistic visualization, but it also incorporates several types of mechanical properties, such as density or modulus of elasticity of the applied materials. The BIM model offers a complete, updated, and accessible database, allowing the data sharing between partners and phases, encompassing all stages of the life cycle of a building [3] [4].

Considering that BIM is relatively recent, the programs are not yet fully prepared, and presents some limitations. In this sense, this paper aims to analyze the technological advances that have been achieved, identifying appropriate strategies for performing different tasks associated with the design of structures. A real building case was worked out in a perspective of modelling a structural design, following the architectural alternative solutions, proving that the communication between both teams is adequate when supported in the BIM technology. In addition, the frequent problems related to the inefficient interoperability that still exist, in the transfer process required along the elaboration of a structural project are identified and strategies of resolution are presented. Finally, the main benefits found on the study are carried out, contributing to the dissemination of BIM application in the structural design [5].

The BIM contribution in the study of alternative proposals, corresponds to support the professionals of the design team, in an easy performance of the eventual adjustments over the model being created, based on a sustained 3D visualization and consultant of the model database, along the creation process. The final design, following the required specifications and adjustments according to the endorsed modifications in a collaborative and integrated way, constitutes to obtain a high-quality product [6]. The improved capability, evident in teamwork, coordination, and collaboration, leads to improved performance of professionals and reduced costs associated with the project design, providing benefits to all involved [7]. In the context of the enterprise, the implementation of BIM requires the restructuring of its internal functional processes, as part of a strategy to adapt traditional working methods to the new paradigm, as well as a change in inter-company communication [8].

Companies have been recognizing that this transition is inevitable and its implementation in work practices should be understood as a mandatory requirement [9]. In all areas of the construction industry, owners, designers, builders, and managers have been reporting the benefits of adopting BIM methodology in their specific activity. This has contributed to the rapid and growing acceptance of BIM, leading government entities to establish guides and deadlines for mandatory implementation in

public buildings [10] [11]. These problems have resulted in a consensus on the need for the BIM approach to be supported by interoperability between software [12]. Currently, interoperability between software points to two main approaches [13]:

Despite the constant effort in research, carried out in academic and business environments, the persistent limitation of interoperability in some of the tasks that require model transfer, in the pursuit of the development of different stages of the project, the technology market still does not provide a completely effective solution, negatively affecting its implementation [14]. This makes interoperability the main practical barrier to successful BIM adoption in the industry [15].

The study presents an evident collaboration between the architect and the structural engineer supported on BIM software. The changes that were made first in the architectural design were then transposed to the structural solution, and several adjustments were made accordingly.

II. CASE STUDY (ARCHITECTURE VS STRUCTURES)

The case study chosen, to develop the structural design in BIM environment, is a single-family house, located in the archipelago of the Azores, Portugal, on the island of Faial. The house has two-bedroom with 258.3 m² of gross floor area and 182.2 m² of building area. The process of creating the architecture design, admitted several alternative options, providing, in the ambit of this study, an illustration of how the BIM model constitutes an adequate work basis and understanding of coordination between the activities of the architect and the structural engineer. The preliminary drawings were provided by the engineering office, involved in the design, which provided the architectural plans and elevations, as well as a preliminary design of the structure, in a CAD format file.

A. Pre-dimensioning proposal

The proposed structural solution consists essentially of a reinforced concrete beam structure, formed by columns, beams, walls and slabs, and a complementary mixed structural system, formed by frames and walls. The modelling of the foundations, in a first phase of support to the architectural study, was only considered the modelling of the structure for calculation purposes, i.e., the representation of the analytical model.

The model shows, at level 0 the adopted structural walls, of 0.20m and 0.25m thickness. On floors 1 and 2, the solid slabs are 0.17m thick, supported by a beamed system (Figure 1).

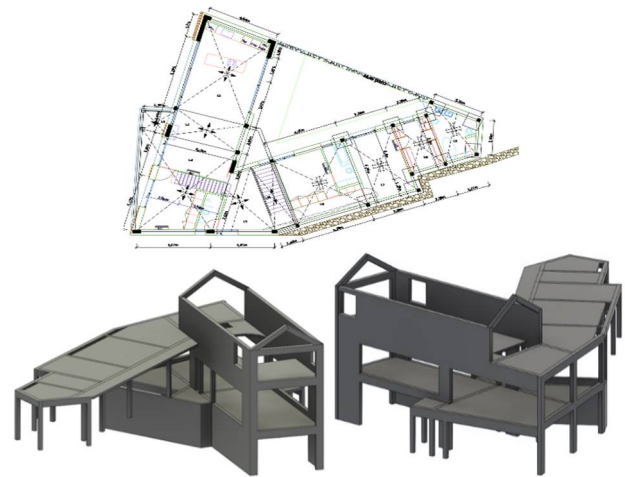


Figure 1. : Imported CAD drawings and perspectives of the BIM model of structures created in the Revit system.

The structural walls conceived for floor 2, corresponds to the structural solution found to meet the architectural constraints. However, in seismic terms, these elements are not recommended, because they introduce more mass in the upper floor and, consequently, increase the horizontal forces.

The building also has a sloped roof, in reinforced concrete slab, on the main body, and an accessible terrace roof, on the secondary body. Both slabs are 0.15m thick. The structural solution presented was the result of a series of adaptations, made to the initial pre-dimensioning proposal, which occurred because of architectural changes made. The model was created based on the drawings of the architectural design that was imported to Revit software.

B. Study of alternative solutions

The project under analysis was submitted to changes made in the architecture design, while the structural model was being developed, forcing a series of adaptations over the structural solution. The most relevant project changes were analysed, and the BIM model of structures were adjusted according to the imposed modifications. The architect proposed some changes to the project, which in several situations led to problems that were difficult for the engineering office to solve (Figure 2):

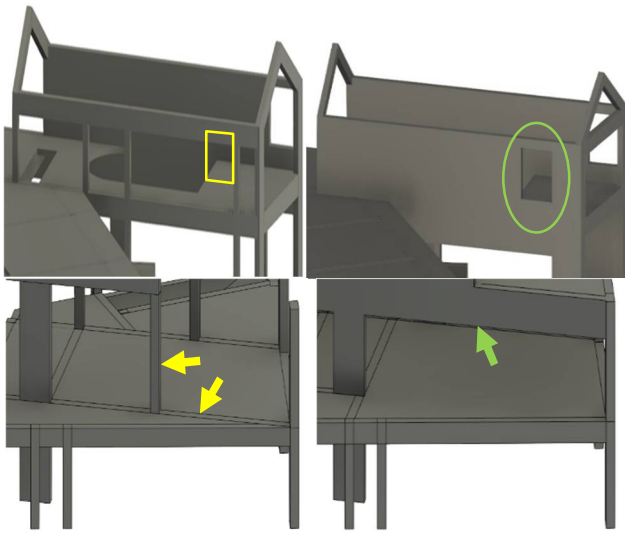


Figure 2. Structural solution before and after the amendments.

- Amendment 1 considered in the model of structures was requested by the office responsible for the architecture, for having reformulated the size of the garage. It was necessary to remove a beam on the 1st floor, as it conditioned the installation of the garage door;
- Amendment 2 requires the placement of the window shown in, making impossible to place a beam with the required dimensions;
- The third amendment required by the architect, consisted of the removal of 2 columns on the 0th floor, creating a contiguous span in almost the entire front of the main body of the building.

These difficulties were often not visually understood by the architectural office, which led to a significant increase in the cost of the building structure, which could have been minimized with minor adjustments in the architectural design. The use of the BIM model to assist the 3D visualization of the project, improved the communication between the offices, even though the architectural office does not use BIM software. Several meetings were made based in architectural drawings and BIM structural model.

The engineering office benefited a lot from the use of BIM in this project, as it was easy and fast to make changes in the structure and perform the respective structural analysis at any stage of the design. If the architectural design had also been executed in BIM, communication will be more easily established, leading naturally to a better understanding, on both sides, of the constraints, requirements, and possibilities of resolution in a collaborative mode.

III. MODEL OF THE STRUCTURAL SOLUTION

A. Geometric and analytical modelling

The geometric and analytical modelling of the structure was made using Revit software. Before modelling, it was important to set up the basic parameters of the structure model to be created, such as design units and materials. After that, modelling begins by defining the project alignments and levels, with the

help of the imported CAD drawings, which serve as the basis for the placement of the structural elements.

Then, the first elements to be modelled are the columns, and sequentially the beams, walls, and slabs. The foundations are modelled after the definition of columns and retaining walls. The modelling process in Revit is very intuitive, supported by the interactive interfaces of the different families of elements available in the system. The user selects an element of the desired family and adjusts the dimensions of its cross section or thickness as designed (Figure 3).

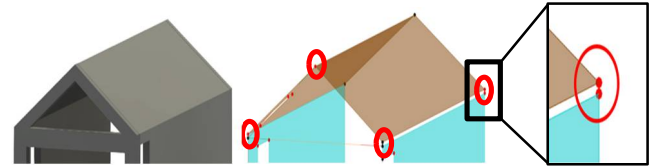


Figure 3. Physical model of the roof and disconnected nodes in the analytical model.

The modelling of some elements, such as corner columns, inverted beams, slabs, and inclined roof elements, displayed some complexity and limitations, mainly regarding the compatibility of geometric and analytical models. The geometric modelling should always be accompanied by the analytical model, which is automatically generated by the software, to detect errors during its definition. These errors can occur when, for example, the geometric limits of the elements do not coincide with the analytical limits or by failures in the program to detect where the analytical connection of the element should be made.

However, there are some tools that facilitate the detection of problems in the analytical model. One of the tools used was a visual filter to detect unconnected analytical nodes. The filter allows, among several options, to highlight disconnected nodes by changing their color to a distinct one, allowing the user to quickly identify them (Figure 4). Changes were also made to the representation of the analytical slabs, allowing a better visualization of the different analytical elements (Figure 4).

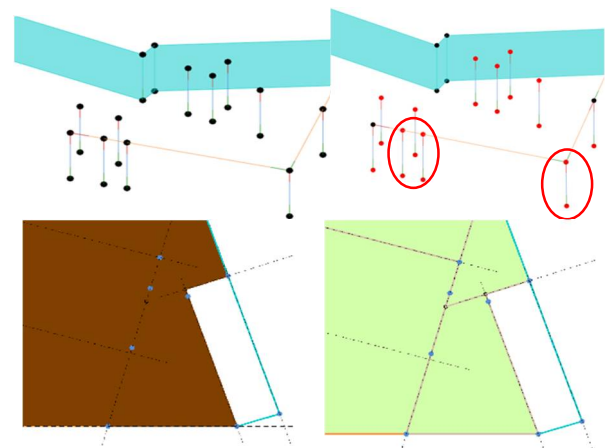


Figure 4. Visualization of the analytical model before and after applying the detect filter and the analytical model of a slab.

B. Structural analysis and design

Once the modelling of the structure has been completed and the consistency of the analytical model has been checked, the

model is transferred to the calculation program, the Robot, in order to perform the structural analysis (Figure 5).

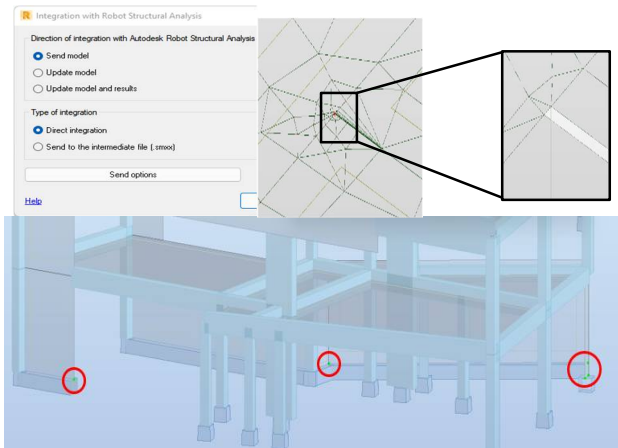


Figure 5. Direct integration option, incoherent zone in the finite element mesh and isolated nodes detected.

The Revit/Robot data transfer of the structural BIM model, between the two BIM-based systems, was carried out by using the Revit/Robot plug-in, through the direct integration option, allowing the rapid two-way transfer of information. No errors were detected when performing the transfer, although some flaws were identified later in the analytical model because of some slab nodes were poorly connected in Revit (Figure 5). Once the model was checked for consistency, the loads and loads combinations were defined:

- The loads considered for the dead loads, were the self-weight, that is automatically defined and applied by the calculation program, plus the remaining loads, such as partition walls and floor coverings. The applied live loads were defined according to the Portuguese norm, NP EN 1991-1-1, classifying the case study in category A with the type of use "Domestic and residential activities". After the definition of dead and live loads, the seismic action was considered through a modal analysis by response spectrum since the building has an irregular geometry and is in a high seismicity zone. Then, the structural analysis was performed (Figure 6);

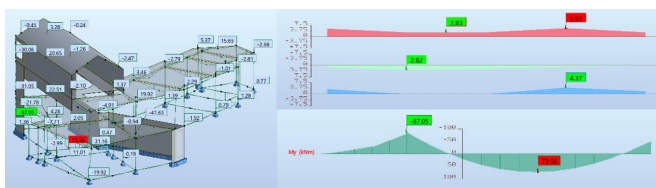


Figure 6. Individual analysis of the beam of alignment A and My moment and required reinforcement areas

- The analysis results are obtain using Robot. The Robot allows a complete structural analysis of the building structure, providing diagrams of efforts, reactions, deformations, and effort maps, among other options. The information can be presented graphically or in tables. The visualization of the results can be global or more detailed, allowing the analysis of the elements individually. It is also possible to present the results of

each load case separately, or to display the envelope forces according to the defined combinations, making it easier to identify critical points. These results can be transferred to Revit and can be consulted;

- Reinforcement detailing: Once the structural analysis was done, the reinforcement for each structural component was calculated. Based on the results of efforts and behaviour of the structure, the Robot software automatically calculates the necessary reinforcement, satisfying the requirements established in the related Eurocode. Subsequently, it generates the correspondent technical drawings of reinforcement detailing.

The calculation of the reinforcement of the elements, except for the reinforcement of slabs and walls, is done through the provided reinforcement plug-in. The reinforcement detailing is rigorous, although some problems and limitations have been detected (Figure 7):

- In foundation design, it is not possible to create combined footings with more than 2 columns;
- The eccentricity applied in Revit is not considered by the Robot, and consequently the calculation of the suspension reinforcement is not correct;
- In the detailing of the inclined roof beams, the reinforcement automatically generated by the program did not verify all the Eurocode conditions, and manual adjustments had to be made;
- Although slab reinforcement can be dimensioned in Robot, it cannot be transferred to Revit;
- The reinforcement design of structural walls has not yet been implemented in Robot according to Eurocode.

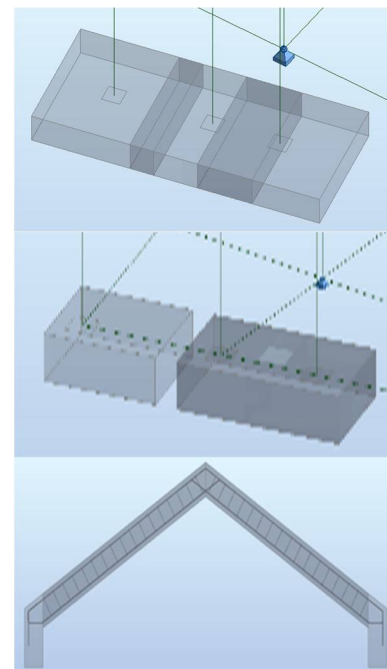


Figure 7. Geometric overlap of the footings and the adopted solution and roof beams.

IV. TRANSFER OF ANALYSIS RESULTS

The analysis of the interoperability capability between Revit and Robot (2022 versions), regarding the reverse transfer process, Robot/Revit, where there is usually a greater limitation, is evaluated in this item. On each type of element, the identified inconsistencies were analyzed, and a possible resolution strategy was presented, in order to overcome the difficulties found. The lack of interoperability demonstrated between the programs, led to the option of redoing some procedures, to "unite" the two models again. A general update was then performed on the Revit model, where most of the elements had to be adjusted, and then the Revit/Robot transfer was performed for the structural calculation. Afterwards, the update operation of the Revit model was performed again, and, in the transfer process, there were no longer any warnings regarding significantly different models:

- The foundation reinforcement transfer from Robot to Revit is only performed if its geometry is previously defined in Revit. Although the transfer is performed, in some situations the reinforcement has a skewed orientation in relation to the footings (Figure 8). This type of errors, although easy to adjust, force the user to spend some time for the correct representation of the reinforcement detailing in the foundation elements;
- After that, the reinforcement transfer improved, however, some elements were not transferred correctly. Analysing the transfer of the columns, no problems were identified. However, the software showed some difficulties in locating the geometry position of the beams and their orientation. Additionally, after the necessary adjustment of the reinforcement, it was verified in some cases that there were more stirrups than those that were necessary for the length of the spans (Figure 8). This happened because, since the Robot does not consider eccentricities, the software considered larger spans than those defined in Revit;
- In the Robot/Revit transfer of the reinforcement of the columns and beams, an obstacle kept appearing, which made it impossible to transfer the information quickly. When attempting to update these elements in Revit, the software reported that the Revit model was significantly different from the Robot model (Figure 8). When proceeding with the update, the elements were simply deleted from the model.

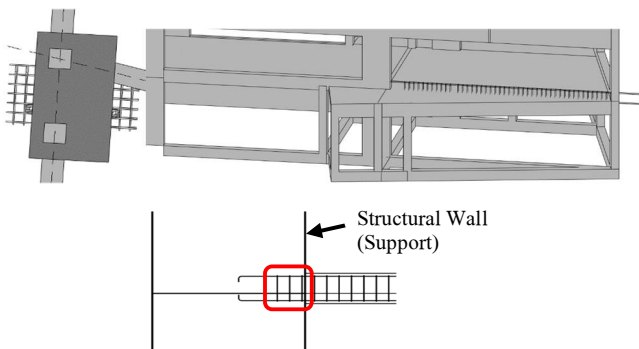


Figure 8. Errors in the transfer reinforcement in foundations and beam elements

The direct integration between Revit and Robot, allows the transfer of the structural analysis results to the Revit model. This capability is especially useful when choosing and modelling slab reinforcement in Revit, where the required reinforcement area in each panel is quickly extracted (Figure 9). Additionally, it is possible to transfer the resulting stresses, displacements, and strains in columns and beams, concerning each loads combination.

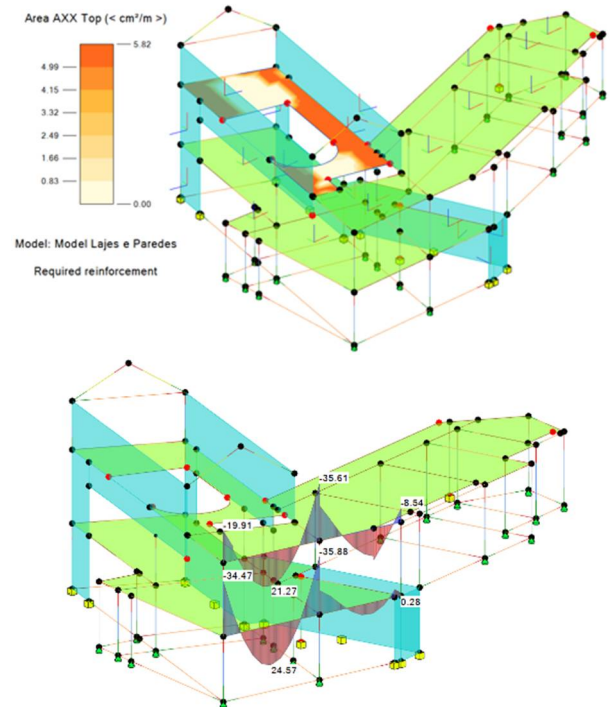


Figure 9. Required reinforcement area in slab floor 2 and in beams.

Despite this integration, demonstrating the ability to centralize the results of the structural analysis in Revit, this process still has several errors. There were some difficulties trying to transfer the results. The only solution found was to perform the update of the model, in a selective way by elements, beams and columns and by groups of elements, such as slabs and walls. This procedure led to better transfer results.

V. CONCLUSIONS

The technological advances associated with the BIM methodology have been showing benefits in productivity and construction quality. This study aimed to analyses the benefits of using BIM methodology in the execution of a structural design, as well as the limitations related to interoperability between Revit and Robot software (2022 versions), and the individual limitations of each program. The main benefits are focused essentially on aspects related to the correct functionality, speed, and simplicity in the execution of some processes:

- The ability to transfer information bidirectional between software, allows changes to the project to be made in only one of the programs, avoiding performing the same task twice, the transfer of the BIM model from Revit to Robot is very reliable and in Robot, the automatic

calculation and reinforcement detailing according to the Eurocodes showed very satisfactory results;

- 3D/BIM modelling, compared to traditional methods, enables any change to the project to be automatically applied to all plans, sections, and views and the possibility to save all definitions, families of elements and annotations, to be reused in future projects, preserving common elements between projects.
- Despite the multiple benefits and technological advances, some limitations were still found:
- It is not possible to transfer the reinforcement of slabs from the Robot to Revit, the transfer to the modelling software of the reinforcement of isolated footings, is only done if its geometry is previously defined in Revit and transferring the reinforcement of wall foundations from Robot to Revit is not possible;
- The use of Revit as the most applied modeller system was also applied in this case, but the most important innovation presented in the study concerns the collaboration architecture/structures. As all changes in the first projects in easily analysed viewing and consulting the BIM model that is absorbing all adjustments.

Regarding the limitations of the study, the inexperience of the author in manipulating the software may have limited the use of advanced tools, with all their potential, which would allow a better use of all the capabilities of the programs. Another noteworthy aspect was the release of new versions of the software close to the date of completion of the study, making it impossible to carry out the study using the most recent versions.

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